Project Summary

LectureIQ — AI-Powered Learning Co-Pilot for Engineering Students

Problem Statement

Over 10 million engineering students in India attend 4–6 hours of daily lectures and spend an additional 10–15 hours per week manually organizing notes. Students in Tier 2 and Tier 3 cities face a compounded challenge: lectures are delivered in Hindi-English code-mixed language, making them difficult to review and study from, while access to quality supplementary resources remains extremely limited. The result is poor retention, inefficient revision, and an unequal education system where geographic location determines academic outcomes.

Our Solution

LectureIQ is a web-based AI platform that converts any lecture audio recording into a complete set of study materials — automatically. A student uploads an MP3/WAV file and receives structured notes, 10–15 flashcards, 5–10 quiz questions with explanations, and curated YouTube/documentation resources — all within 3 minutes. The platform also tracks quiz performance, identifies weak topics, and uses spaced repetition to optimize revision schedules.

Live Application: https://lecture-iq-seven.vercel.app

Demo Video: https://youtu.be/h5exUUc-B5w

GitHub: https://github.com/kaustubh05-prog/LectureIQ

Why AI is Required

AI is not a convenience in LectureIQ — it is the **only feasible way** to deliver this solution:

1. **Multilingual Speech Recognition:** Hindi-English code-mixed speech cannot be processed by rule-based systems. OpenAI Whisper's multilingual neural model supports 99 languages and handles code-mixing natively, achieving 85%+ Word Error Rate accuracy on real lecture audio.
2. **Contextual Understanding at Scale:** Extracting key concepts, definitions, and theorems from an unstructured lecture transcript requires deep language understanding. We use Groq's LLaMA 3.1 70B — capable of generating pedagogically structured notes, coherent flashcards, and well-reasoned MCQs with explanations — something no manual or rule-based pipeline can replicate.
3. **Speed:** Processing a 1-hour lecture takes a human 3–5 hours. LectureIQ completes the full pipeline — transcription, generation, and resource linking — in under 3 minutes.
4. **Semantic Resource Matching:** spaCy NLP extracts the key topics from a transcript and maps them to relevant YouTube tutorials and documentation links automatically, across any STEM subject.

How AWS Services Are Used

AWS S3 (Amazon Simple Storage Service) is the primary storage backbone of LectureIQ:

- All uploaded lecture audio files are stored in an S3 bucket with **server-side AES-256 encryption**, ensuring student data remains private and secure.
- File access is controlled through **time-limited pre-signed URLs** (1-hour validity), so audio files are never publicly exposed.
- **Lifecycle policies** automatically delete audio files after 30 days, minimizing storage costs and protecting user privacy.
- The architecture is fully optimized within the **AWS Free Tier** (5GB storage, 20K GET, 2K PUT requests/month), keeping total infrastructure cost under $5 for the hackathon period.
- The design is production-ready for scaling to **AWS CloudFront CDN** and **AWS Lambda** for serverless processing in future phases.

Value the AI Layer Adds to User Experience

Without LectureIQ	With LectureIQ
10–15 hours/week on manual notes	Notes ready in under 3 minutes
Incomplete, unstructured notes	Structured markdown with LaTeX formulas
No flashcards unless made manually	10–15 AI-generated flashcards per lecture
No quizzes until exam time	Instant MCQs with detailed explanations
Manually searching for resources	Top 3–5 curated resources auto-linked
No visibility into weak areas	Quiz analytics + weak topic identification

The AI layer transforms passive audio into an **active, personalized learning experience** — democratizing quality education for students who don't have access to coaching, tutors, or premium study material platforms. LectureIQ is completely free for students, built on a zero-cost infrastructure stack, and designed to scale to 10+ million engineering students across India.

Technology Stack Summary

- **Frontend:** React 18 + Vite → Vercel
- **Backend:** Python FastAPI → Render
- **Storage:** AWS S3 (audio) + Supabase PostgreSQL (data)
- **AI/ML:** OpenAI Whisper (transcription) + Groq LLaMA 3.1 70B (generation) + spaCy (NLP) + YouTube Data API v3 (resources)
- **Auth:** JWT + bcrypt
- **Processing:** FastAPI BackgroundTasks with async pipeline